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QUARTERLY EXAMINATION - 2026	12 - STD	
MATHEMATICS	Marks 90	Time 3.00 Hrs.

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PART - I

I. Answer all the questions. **20 x 1 = 20**

1. If $adj\ A = \begin{bmatrix} 2 & 3 \\ 4 & -1 \end{bmatrix}$ and $adj\ B = \begin{bmatrix} 1 & -2 \\ -3 & 1 \end{bmatrix}$ then $adj(AB)$ is

   (a) $\begin{bmatrix} -7 & -1 \\ 7 & -9 \end{bmatrix}$ (b) $\begin{bmatrix} -6 & 5 \\ -2 & -10 \end{bmatrix}$ (c) $\begin{bmatrix} -7 & 7 \\ -1 & -9 \end{bmatrix}$ (d) $\begin{bmatrix} -6 & -2 \\ 5 & -10 \end{bmatrix}$

2. The principal argument of $(\sin 40^0 + i \cos 40^0)5$ is

   a) -110^0 b) -70^0 c) 70^0 d) 110^0

3. If $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$, then the value of x is

   a) 4 b) 5 c) 2 d) 3

4. Area of the greatest rectangle inscribed in the elipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

   a) $2ab$ b) ab c) $\sqrt{ab}$ d) $\frac{a}{b}$

5. If $A^T A^{-1}$ is symmetric then $A^2 =$

   a) A^{-1} b) $(A^T)^2$ c) A^T d) $(A^{-1})^2$

6. If f and g are polynomials of degree m and n respectively and if $h(x) = (f \circ g)(x)$, then the degree of h is

   a) mn b) $m + n$ c) m^n d) n^m

7. If $|Z - 2 + i| \leq 2$ then the greatest value of $|Z|$ is

   a) $\sqrt{3} - 2$ b) $\sqrt{3} + 2$ c) $\sqrt{5} - 2$ d) $\sqrt{5} + 2$

8. If $\vec{a}.\vec{b} = \vec{b}.\vec{c} = \vec{c}.\vec{a} = 0$ then the value of $\left|\vec{a}\ \vec{b}\ \vec{c}\right|$ is

   a) $|\vec{a}||\vec{b}||\vec{c}|$ b) $\frac{1}{3}|\vec{a}||\vec{b}||\vec{c}|$ c) 1 d) -1

9. The radius of the circle $3x^2 + by^2 + 4bx - 6by + b^2 = 0$ is

   a) 1 b) 3 c) $\sqrt{10}$ d) $\sqrt{11}$

10. If $A = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$ and $A(adj\ A) = \begin{bmatrix} K & 0 \\ 0 & K \end{bmatrix}$ then K

    a) 0 b) $\sin\theta$ c) $\cos\theta$ d) 1

11. $\cos(\cos^{-1}x) = x$

    a) $\frac{-\pi}{2} \leq x \leq \frac{\pi}{2}$ b) $0 < x \leq \pi$ c) $0 \leq x \leq \pi$ d) $-1 \leq x \leq 1$

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12. If $y = mx + c$ is a tangent to the parabola $y^2 = 4ax$ then the point of contact

a) $\left(\frac{a}{m^2}, \frac{2a}{m}\right)$ b) $\left(\frac{-a}{m^2}, \frac{2a}{m}\right)$ c) $\left(\frac{a}{m^2}, \frac{-2a}{m}\right)$ d) $\left(\frac{-a}{m^2}, \frac{-2a}{m}\right)$

13. The number of positive zero of the polynomial $\sum_{r=0}^{n} n_{c_r} (-1)^r x^r$ is

a) 0 b) n c) $< r$ d) r

14. If ρ represent the rank and A and B are $n \times n$ matrices, then

a) $\rho(A + B) = \rho(A) + \rho(B)$ b) $\rho(AB) = \rho(A)\,\rho(B)$

c) $\rho(A - B) = \rho(A) - \rho(B)$ d) $\rho(A + B) < n$

15. The coordinates of the point where the line $\vec{r} = (6\hat{i} - \hat{j} - 3\hat{k}) + t(-\hat{i} + 4\hat{k})$ meets the plane $\vec{r}.(\hat{i} + \hat{j} - \hat{k}) = 3$ are

a) (2, 1, 0) b) (7, -1, -7) c) (1, 2, -6) d) (5, -1, 1)

16. $cis\frac{28\pi}{5} = 6\pi - \frac{2\pi}{5}$

a) $cis\left(\frac{-2\pi}{5}\right)$ b) $cis\left(\frac{2\pi}{5}\right)$ c) $cis\left(\frac{3\pi}{5}\right)$ d) $cis\left(\frac{-3\pi}{5}\right)$

17. If the direction cosines of a line are $\frac{1}{c}, \frac{1}{c}, \frac{1}{c}$ then

a) $c = \pm 3$ b) $c = \pm\sqrt{3}$ c) $c > 0$ d) $0 < c < 1$

18. If $x + y = K$ is a normal to the parabola $y^2 = 12x$ then the value of K is

a) 3 b) -1 c) 1 d) 9

19. If the coordinates at one end of a diameter of the circle $x^2 + y^2 - 8x - 4y + c = 0$ are (11, 2) the coordinate of the other end are

a) $(-5, 2)$ b) $(-3, 2)$ c) $(5, -2)$ d) $(-2, 5)$

20. The shortest distance between the two skew lines $\vec{r} = \vec{a} + s\vec{u}$ and $\vec{r} = \vec{b} + t\vec{v}$ is

a) $\frac{|(\vec{b} - \vec{a}).(\vec{u} \times \vec{v})|}{|\vec{u} \times \vec{v}|}$ b) $\frac{|(\vec{b} - \vec{a}).(\vec{u} \times \vec{v})|}{\vec{u} \times \vec{v}}$ c) $\frac{|(\vec{b} - \vec{a}).(\vec{u} \times \vec{v})|}{|\vec{a} \times \vec{b}|}$ d) $\frac{|(\vec{b} - \vec{a}).(\vec{u} \times \vec{v})|}{|\vec{a}|}$

PART - II

Answer any seven questions. Q.No. 30 is compulsory. **7 x 2 = 14**

21. Sove by cramer's rule the system of equation $\frac{3}{x} + 2y = 12, \frac{2}{x} + 3y = 13$

22. find the value of $\sec^{-1}\left(\frac{-2\sqrt{3}}{3}\right)$

23. Find the vertices, foci for the hyperbola $9x^2 - 16y^2 = 144$.

24. Prove that $\left[\vec{a} - \vec{b}, \vec{b} - \vec{c}, \vec{c} - \vec{a}\right] = 0$

25. If $\omega \neq 1$, is a cube root of unity. Show that $\frac{a + b\omega + c\omega^2}{b + c\omega + a\omega^2} + \frac{a + b\omega + c\omega^2}{c + a\omega + b\omega^2} = -1$

26. Find the value of x such that $-5\pi \leq x \leq 5\pi$ and $\cos x = 1$

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27. Find the rectangular form of the complex number $\frac{\cos\frac{\pi}{6}-i\sin\frac{\pi}{6}}{2(\cos\frac{\pi}{3}+i\sin\frac{\pi}{3})}$

28. $\vec{a}=-3\hat{i}-\hat{j}+5\hat{k}, \vec{b}=\hat{i}-2\hat{j}+\hat{k}, \vec{c}=4\hat{j}-5\hat{k}$ find $\vec{a}.(\vec{b}\times\vec{c})$

29. If K is real, discuss the nature of the roots of the polynomial equation $2x^2+Kx+K=0$

30. If $(AB)^{-1}=\begin{bmatrix}12 & -17\\ -19 & 27\end{bmatrix}$ and $A^{-1}=\begin{bmatrix}1 & -1\\ -2 & 3\end{bmatrix}$ find B^{-1}

PART - III

III. Answer any seven questions. Q.No. 40 is compulsory. **7 x 3 = 21**

31. If $\frac{Z+3}{Z-5i}=\frac{1+4i}{2}$ find rectangular of Z.

32. Find the rank of the matrix $\begin{bmatrix}4 & -2 & 6 & 8\\ 1 & 1 & -3 & -1\\ 15 & -3 & 9 & 21\end{bmatrix}$

33. Solve : $7x^3-43x^2=43x-7$

34. Show that $\tan^{-1}x+\tan^{-1}y+\tan^{-1}z=\tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$

35. If $y=2\sqrt{2}x+c$ is a tangent to the circle $x^2+y^2=16$, find the value of c.

36. Find acute angle between the planes $\vec{r}.(2\hat{i}+2\hat{j}+2\hat{k})=11$ and $4x-2y+2z=15$

37. Solve the equation $2\cos^2 x - 9\cos x + 4 = 0$

38. Find the magnitude and the direction cosines of the torque about the point (2, 0, -1) of a force $2\hat{i}+\hat{j}-\hat{k}$, whose line of action passes through the origin.

39. If $\cot^{-1}\left(\frac{1}{7}\right)=\theta$ then find the value of $\operatorname{cosec}\theta$

40. Length of semi mirror of the elliptical past of an optical lens system is 3 cm. The parabolic part of the system has a focus in common with the right focus of the ellipse. The vertex of the parabola is at the origin and the equation of parabola $y^2=4\sqrt{7}x$ opens to the right. Determine equation of the ellipse and also centre of the ellipse at the origin.

PART - IV

JV. Answer all the questions. **7 x 5 = 35**

41. a) Investigate for what values of λ and μ the system of linear equations $x+2y+z=7, x+y+\lambda z=\mu, x+3y-5z-5$ has (i) no solution (ii) a unique solution (iii) an infinite number of solutions.

(OR)

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b) A semielliptical archway over a one-way road has a height of 3 m and a width of 12 m. The truck has a width of 3m and height of 2.7 m. Will the truck clear the opening of the eachway?

42. a) Find the coordinates of the foot of the perpendicular and length of the perpendicular from the point (4, 3, 2) to the plane $x + 2y + 3z = 2$

(OR)

b) Two coast guard stations are located 600 km apart at points $A(0, 0)$ and $B(0, 600)$. A distress signal from a ship at P is received at slightly different times by two stations. It is determined that the ship is 200 km farther from station A than it is from station B. Determine the equation of hyperbola that passes through the location of the ship.

43. a) If $Z = x + iy$ is a complex number such that $\left|\frac{Z-4i}{Z+4i}\right| = 1$ show that the locus of Z is real axis.

(OR)

b) Discuss the maximum possible number of positive and negative zeros of the polynomials $x^2 - 5x + 6$ and $x^2 - 5x + 16$. Also draw rough sketch of the graphs.

44. a) Solve : $\tan^{-1}(x-1) + \tan^{-1}x + \tan^{-1}(x+1) = \tan^{-1}(3x)$

(OR)

b) Find the vertex, focus, directrix and length of the latus rectum of the parabola $x^2 - 4x - 5y - 1 = 0$.

45. a) State and prove Apollonius's theorem by vector method.

(OR)

b) Find all cube roots of $\sqrt{3} + i$

46. a) If $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$, verify that $A(adj\ A) = (adj\ A)\ A = |A|I_3$

(OR)

b) If $\cos^{-1}x + \cos^{-1}y + \cos^{-1}z = \pi$ and $0 < x, y, z < 1$, show that $x^2 + y^2 + z^2 + 2xyz = 1$.

47. a) Solve the equation $6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0$ if its is known that $\frac{1}{2}$ is a solution.

(OR)

b) Show that the lines $\frac{x-3}{3} = \frac{y-3}{-1}, z-1=0$ and $\frac{x-6}{2} = \frac{z-1}{3}, y-2=0$ intersect. Also find the point of intersection.

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